amount of memory used and data transferred are reduced. *H_2O* (Zhang et al., 2023), *Scissorhands* (Liu et al., 2023a) and *FastGen* (Ge et al., 2024) are examples of such eviction methods. H_2O uses a greedy eviction policy that maintains in memory the most salient "Heavy Hitter" tokens that contribute most to the attention scores. Scissorhands identifies and maintains "pivotal tokens" by counting when a token's attention score exceeds an importance threshold. FastGen adopts heuristics such as preventing the eviction of special tokens and punctuation, and tailors the compression strategy to each individual attention head. While these methods reduce the memory footprint of the KV cache as well as data transfer, they also lead to permanent loss of information from the context window, which can lead to mistakes for queries seeking less-attended parts of the sequence.

IceFormer (Mao et al., 2023) uses multiple existing approximate nearest neighbour algorithms for approximating attention scores of pre-trained models, focusing on speeding-up the prefill stage, rather than generation. *Scatterbrain* (Chen et al., 2021) employs similar techniques, but for computer vision applications.

In addition to compressing the KV cache, a number of methods strive to speed up LLM inference by inducing sparsity in the weights of the model. Deja Vu (Liu et al., 2023c) is a contextually-sparse approach that aims to predict which model parameters are required such that the error between the full computation and sparse approximation is minimised. Similarly, activation sparsity methods, including Kurtz et al. (2020) and Mirzadeh et al. (2024), exploit zero-values found in activations, typically induced by ReLU activation functions. Kurtz et al. (2020) introduce an alternative *forced activation threshold ReLU* function which can induce sparsity at specified thresholds. Similarly, Mirzadeh et al. (2024) replace the activation functions in LLMs with ReLUs, followed by additional fine-tuning. These methods are most suitable for small batch size and short sequence length regimes, where inference is bottlenecked by parameter transfer, rather than the KV cache, but are compatible with sparse attention techniques such as SparQ Attention.

An orthogonal line of work increases bandwidth efficiency by compressing the KV cache with 4-bit number formats (Liu et al., 2023b; Sheng et al., 2023). Liu et al. (2023a) demonstrate that 4-bit compression is complementary to techniques that reduce the number of transferred elements.

8. Conclusion

In this work we have presented SparQ Attention, a novel technique for unlocking faster inference for pre-trained LLMs. Our proposed technique modifies the attention mechanism to access only the relevant tokens from the KV cache at every generation step, leading to considerable data transfer savings. This is particularly beneficial in long sequence length regimes, where inference speed is often bottlenecked by memory transfers rather than computation.

We also highlight the advantages of maintaining the full KV cache in memory for task performance by comparing SparQ Attention to other popular strategies which discard information from the input sequence. These alternative approaches rely on heuristics or predefined policies to determine which items in the KV cache to remove, which may not generalise across the wide range of applications LLMs are used for. We show that SparQ Attention is robust across numerous tasks and models, making it a viable technique for reducing inference times in unseen settings.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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